

Lesson 5 — Design Challenge — Build-a-Bot Parade

Matches LEGO Teacher Portal lesson: "Pet Sitter".

Objective (I can...): I can plan, build, and code a little machine that does steps I chose, in order. **CS concept:** Putting order, little steps, and a plan together · **Builds on:** Lessons 1–4. *Also serves as flex/catch-up — if earlier lessons ran long, use today to finish and celebrate.*

AI Connection (2 min, read aloud): Making an AI toy is the same loop you'll use today: break it down → plan → build → try it. *Ask the group:* "Which of those steps do you think people redo the most?"

Materials: LEGO CS & AI kit + device per pair, plan mats, a simple picture challenge card. **Model already built** (Basics model from Lesson 2) — so the 25 minutes is plan + code + play, not build-from-scratch. Pairs may add small bricks only.

60-minute flow - Warm-up (10): Review song/carousel — fast call-and-response: "What's a program?" "Steps in order!" "What's breaking it down?" "Little steps!" - **Concept + demo (8):** Show the open challenge: make the bot do 3+ steps in an order you choose. Coach says a plan for one idea, then hands off. - **Build/Code (25):** Pairs plan and code a 3+ step program on the built model (small brick changes allowed). Coach cheers and nudges, doesn't solve. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add one "surprise" step (light, sound, or extra move) and add it to the plan. - **Share + tidy-up (10):** **Build-a-Bot parade** — bots march on their trays in a gallery walk; each pair names the one thing that helped most. **Coach photographs each build for the unit parent note.**

Pair roles: Co-plan, then Driver codes / Navigator points to the plan; swap at the wiggle break. (*Trio rotates plan/code/test.*)

Differentiation: *Support* — a 2-step target with a partial plan to finish. *Extension / fast-finisher* — require 5 steps with a light and a sound. *Reteach* — hand them their Lesson 4 plan as a starting template.

Exit check: Pair shows a working 3+ step program with a matching plan and can tell what it does.

Progress layer: Send home the **Unit 1 parent note** ("your child can plan and code steps in order") with the **build photo**. **At-home (optional):** Draw a dream robot and label 3 things it does in order. ~5 min.